Isometric Immersions of Negatively Curved Metrics

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Hilberts Theorem

Theorem (Hilbert 1901)

There is no isometric immersion of a complete, smooth surface of constant negative curvature into $\mathbb{R}^3$.

Overview

1. Classical Surface Geometry
2. Codazzi Pairs
3. Hilberts Theorem
4. Extensions of Hilberts Theorem

Section 1

Classical Surface Geometry

Curvature

What is curvature?

Curvature

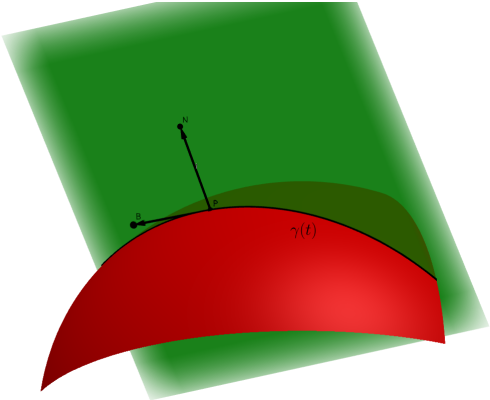

Figure: Normal section of a curved surface

First and Second Fundamental Form

Definition (First Fundamental Form)

A surface in $\mathbb{R}^n$ inherits an inner product which we call the *first fundamental form* and denote I.

Also associated to a surface in $\mathbb{R}^3$ is:

- The *second fundamental form* $II_p : T_pM \times T_pM \to \mathbb{R}$ - symmetric, non-degenerate.
- The *shape operator* S satisfying $I(SX, Y) = II(SX, Y)$ - self adjoint.

Shape Operator and Second Fundamental Form

In coordinates, we can write the metric I and second fundamental form II in matrix form as

$$I = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \quad II = \begin{pmatrix} e & f \\ f & g \end{pmatrix}$$

Remark

The shape operator S can be expressed as $I^{-1}II$ in coordinates.

Extrinsic Curvature

Definition

The *extrinsic* curvature K, is defined to be

$$K = \frac{eg - f^2}{EG - F^2} = \kappa_1 \kappa_2 = \det(S)$$

where κ_1, κ_2 are the eigenvalues of S.

This is Intrinsic!

Theorem Egregium (Gauss 1827)

For a surface in $\mathbb{R}^3$, the quantity $K = \kappa_1 \kappa_2$ is intrinsic!

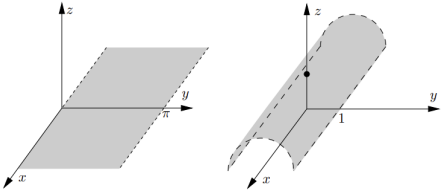

Figure: Two isometric surfaces with different principle curvatures (Lee)

What about surfaces not in $\mathbb{R}^n$?

Riemannian Manifold

Definition

A Riemannian manifold $(M, \langle \cdot, \cdot \rangle)$ is a differentiable manifold M with a smooth $\mathbb{R}$-bilinear form

$$\langle \cdot, \cdot \rangle : TM \times TM \to \mathbb{R}$$

such that $\langle \cdot, \cdot \rangle$ is symmetric and positive-definite.

Isometric Immersion

Definition

An isometric immersion between two Riemannian manifolds (M, g_1) and (N, g_2) is a differentiable map

$$i : N \to M$$

such that the derivative di is injective and $g_2(v, w)|_p = g_1(di(v), di(w))|_{i(p)}$ for all $v, w \in T_pN$ and $p \in N$.

Connections

Definition

Let $(M, \langle \cdot, \cdot \rangle)$ be a Riemannian manifold. A connection on the tangent bundle TM is a map

$$\nabla : \mathrm{Vect}(M) \times \mathrm{Vect}(M) \to \mathrm{Vect}(M)$$

that satisfies $\nabla_X fY = X(f)Y + f\nabla_X Y$ for $f \in C^\infty(M)$ and otherwise is linear.

Connections

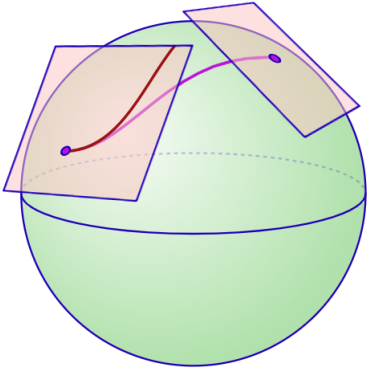

Figure: Connecting two tangent spaces on the sphere (Wikipedia)

The Levi Civita Connection

Definition

On a Riemannian manifold (M, g), there is a unique connection called the Levi Civita connection that is compatible with the metric e.g. the metric g "looks the same" everywhere on M.

Geodesics

Definition

A geodesic is a curve that locally minimizes distances.

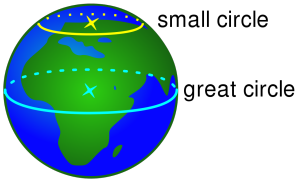

Figure: Great circles are geodesics on the sphere

Completeness

Definition

A Riemannian manifold is said to be complete if every geodesic can be traversed along for an infinite amount of time.

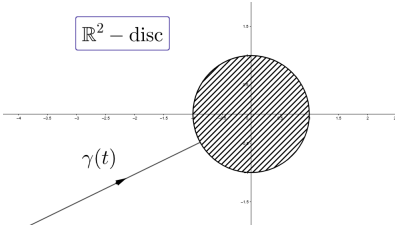

Figure: An incomplete geodesic

Riemann Curvature Tensor

Definition

Let (M, g) be a Riemannian manifold. The Riemann curvature tensor is defined as

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

Remark

This object encodes the intrinsic curvature of a Riemannian manifold.

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Sectional Curvature

Definition

The *sectional curvature* of $\Pi \subset T_pM$ is defined as

$$sec(\Pi) = \langle R(e_1, e_2)e_1, e_2 \rangle$$

where $\Pi = \text{span}(e_1, e_2)$.

Space Forms

Definition

A *space form* is a complete Riemannian manifold M with constant sectional curvature.

Remark

If M is a complete simply connected Riemannian manifold with constant sectional curvature c

- $c = 0 \implies M \cong \mathbb{R}^n$
- $c = 1 \implies M \cong S^n$
- $c = -1 \implies M \cong \mathbb{H}^n$

Section 2

Codazzi Pairs

Gauss Equation

Let $\mathbb{M}^3(c)$ denote the unique simply connected space form with sectional curvature c.

Definition

The Gauss equation of an immersion $f : \Sigma \to \mathbb{M}^3(c)$

$$K(I) = K + c$$

where $K = \det(S)$ is the extrinsic curvature and $K(I)$ is the intrinsic curvature e.g. the sectional curvature $sec(T_p\Sigma)$.

Codazzi Equation

Definition

The Codazzi equation associated to the pair (I, II) with shape operator S is given by

$$\nabla_X SY - \nabla_Y SX = S[X, Y]$$

Bonnet Theorem

Definition

Let $U \subset \mathbb{R}^2$ be open and let I be a Riemannian metric on U and II be a symmetric bilinear form on U. If I and II satisfy the Gauss-Codazzi equations then there is an immersion

$$i : U \to \mathbb{R}^2$$

such that I is the Riemannian metric and II is the second fundamental form.

Fundamental Pair

Definition

A fundamental pair on an oriented surface Σ, is a pair of real quadratic forms (I, II), such that I is a Riemannian metric

Remark

Associated to a fundamental pair is a shape operator S defined by

$$II(X,Y) = I(SX,Y), \quad X,Y \in \mathit{Vec}(\Sigma)$$

Codazzi Pair

Definition

A fundamental pair (I, II) with shape operator S is a Codazzi pair if the Codazzi equation is satisfied - i.e.

$$\nabla_X SY - \nabla_Y SX - S[X, Y] = 0$$

for $X, Y \in \mathrm{Vec}(\Sigma)$ and ∇ is the induced Levi-Civita connection on Σ.

Section 3

Hilberts Theorem

Hyperbolic Space

Definition

Hyperbolic space, $\mathbb{H}^n$ is the unique simply connected, complete Riemannian manifold of constant negative sectional curvature -1.

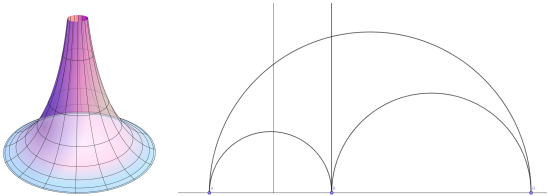

Figure: Pseudo-sphere and Poincaire half plane model of the hyperbolic plane $\mathbb{H}^2$

Asymptotic Curves

Definition

An asymptotic direction $v \in T_p\Sigma$ on Σ, is a direction where $II_p(v, v) = 0$

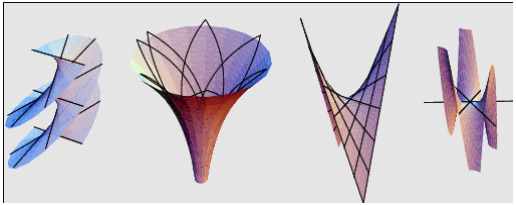

Figure: Asymptotic curves on some surfaces for the second fundamental form. (Wolfram)

Asymptotic Curves

Lemma

If the extrinsic curvature K satisfies $K < 0$ at a point $p \in \Sigma$, then there are exactly 2 asymptotic directions.

Remark

By paramaterizing our surface locally by curves tangent to these directions we can arrange that $e = g = 0$ and $f > 0$ in the matrix of II.

Arc Length Paramaterization

Proposition (José A. Gálvez)

Let (I, II) be a fundamental pair on an oriented, simply connected surface of negative constant extrinsic curvature K. Then we can write (I, II) as

$$I = du^2 + 2\cos\omega du dv + dv^2, \quad II = 2\sqrt{-K}\sin\omega du dv$$

with $0 < \omega < \pi$ as $1 - F^2 > 0$.

Change of Coordinates

Paramaterizing by curves along the asymptotic directions

$$I = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \quad II = \begin{pmatrix} 0 & f \\ f & 0 \end{pmatrix}, \quad f > 0$$

- If $E_v = G_u = 0$ then we can paramaterize by arc length as $E = E(u)$ and $G = G(v)$
- This will hold true if $\nabla_{\frac{\partial}{\partial u}} \frac{\partial}{\partial v} = \nabla_{\frac{\partial}{\partial v}} \frac{\partial}{\partial u} = 0$

Can we parametrize nicely?

Letting $D = EG - F^2$ we can calculate the following conditions on the Christoffel symbols using Koszul's formula

$$\frac{D_u}{2D} = \Gamma^1_{11} + \Gamma^2_{12}, \quad \frac{D_v}{2D} = \Gamma^2_{22} + \Gamma^1_{12} \tag{1}$$

Now using the Codazzi equation we get the following condition on the Christoffel symbols

$$\frac{f_u}{f} = \Gamma^1_{11} - \Gamma^2_{12}, \quad \frac{f_v}{f} = \Gamma^2_{22} - \Gamma^1_{12} \tag{2}$$

Can we parametrize nicely?

Subtracting the corresponding equations of $(1) - (2)$ we obtain

$$\frac{f_u}{f} - \frac{D_u}{2D} = 2\Gamma^2_{12}, \quad \frac{D_v}{2D} - \frac{f_v}{f} = 2\Gamma^1_{12} \tag{3}$$

We now have that $K = -\dfrac{f^2}{D}$ is a negative constant $\implies K_u = K_v = 0$.

Remark

These derivatives give exactly the LHS and RHS respectively and so $\Gamma^2_{12} = \Gamma^1_{12} = 0$

Arc Length Paramaterization

Proposition (José A. Gálvez)

Since $E(u,v) = E(u)$ and $G(u,v) = G(v)$ we can parametrize by arc length and write our metric as

$$I = du^2 + 2\cos\omega du dv + dv^2, \quad II = 2\sin\omega du dv$$

with $0 < \omega < \pi$ as $1 - F^2 > 0$.

We have a global diffeomorphism!

Proposition (José A. Gálvez)

Let $III = du^2 - 2\cos\omega du dv + dv^2$. Then $I + III = 2(du^2 + dv^2)$ is a complete flat metric when I is complete and

$$(u, v) : (\Sigma, du^2 + dv^2) \to \mathbb{R}^2$$

is a local isometry, hence a covering map by completeness and hence a global isometry by simply connectedness. (TODO: Add Sources)

Hilberts Theorem

Theorem (José A. Gálvez, Hilbert)

Let Σ be a complete surface of constant negative Gauss curvature $K(I)$. Then there is no isometric immersion

$$f : \Sigma \to \mathbb{M}^3(c)$$

for $c = 0, 1$ and if $c = -1$ there is no such immersion if $K(I) < -1$.

Hilberts Theorem Proof

Proof.

By our first theorem we can find global coordinates (u, v) defined in $\mathbb{R}^2$ such that

$$I = du^2 + 2\cos\omega du dv + dv^2,$$

$$II = 2\sqrt{-K}\sin\omega du dv$$

and $0 < \omega < \pi$. The Gauss equation *requires* that

$$\omega_{uv} = -(K + c)\sin\omega$$

where $-(K + c) > 0$. □

Section 4

Extensions of Hilberts Theorem

Non-constant negative curvature

Theorem (Efimov 1963)

Let Σ be a complete surface with Gauss curvature $K(I) \leq \delta < 0$. Then there is no isometric immersion

$$f : \Sigma \to \mathbb{R}^3$$

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Strictly Seperated Curvatures

Definition

The principle curvatures, k_1, k_2, of a fundamental pair (I, II) are called *strictly separated* if there exists numbers c_1, c_2 such that

$$k_1 \leq c_1 < c_2 \leq k_2$$

Non-constant negative curvature

Theorem (José A. Gálvez, Antonio Martínez, José L. Teruel)

Let (I, II) be a Codazzi pair on Σ with strictly separated principal curvatures. If (Σ, I) is a complete surface with Gauss curvature $K(I) \leq 0$ and I is not flat then Σ is homeomorphic to a plane and

$$\int_\Sigma |K(I)| dA \leq 2\pi$$

where dA is the area form induced by I.

Non-constant negative curvature

Corollary (José A. Gálvez, Antonio Martínez, José L. Teruel)

Let $\varphi : \Sigma \to \mathbb{H}^3$ be an immersion with Gaussian curvature $K(I) \leq -1$, and one of its principle curvature functions satisfying

$$k_i^2 \geq \epsilon^2 > 0, \quad \text{for } \epsilon > 0$$

then φ is not a complete immersion.

Questions?

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